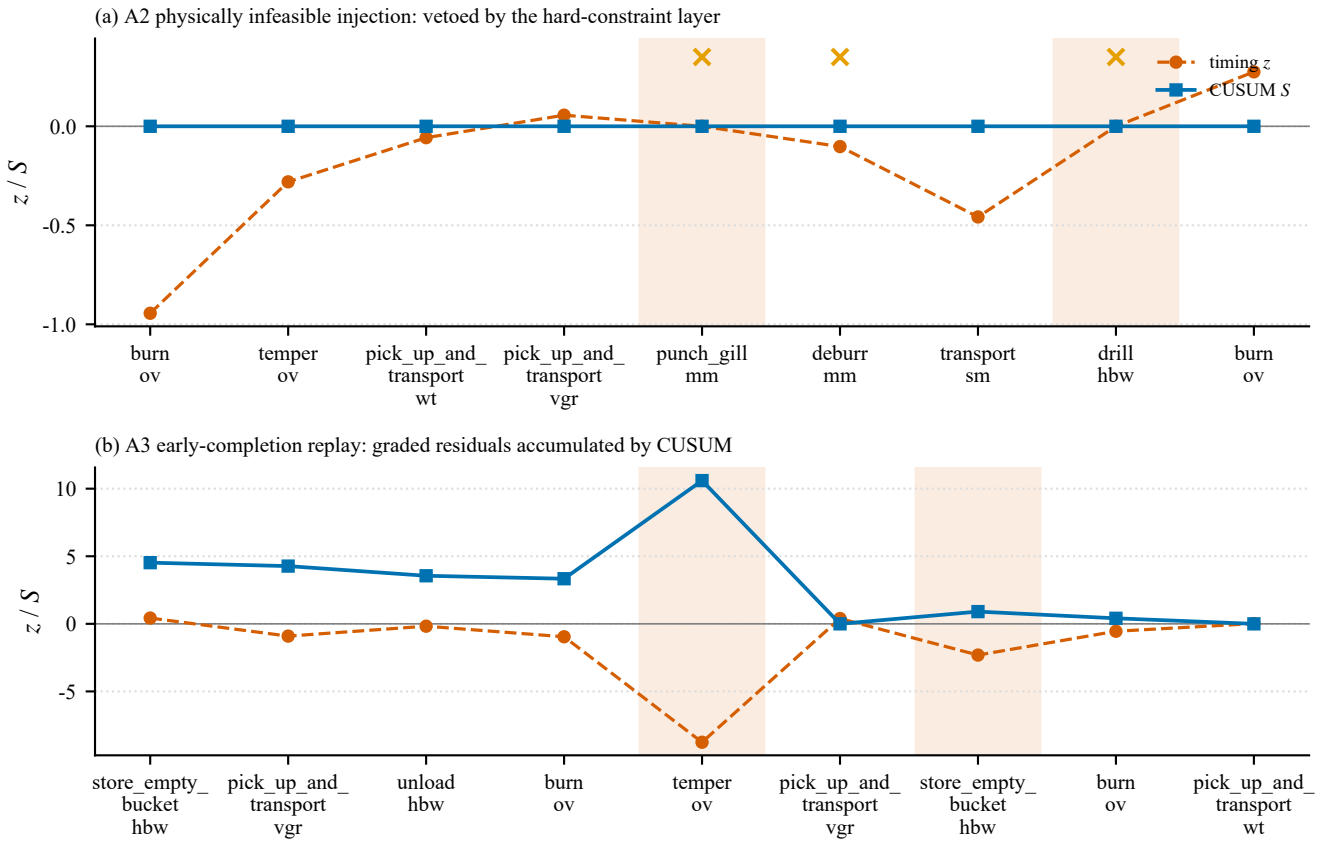

(a) A2 physically infeasible injection: vetoed by the hard-constraint layer
timing z
CUSUM S
z / S
0.0
-0.5
-1.0
burn ov
temper ov
pick_up_and_ transport wt
pick_up_and_ transport vgr
punch_gill mm
deburr mm
transport sm
drill hbw
burn ov
(b) A3 early-completion replay: graded residuals accumulated by CUSUM
z / S
10
5
0
-5
store_empty_ bucket hbw
pick_up_and_ transport vgr
unload hbw
burn ov
temper ov
pick_up_and_ transport vgr
store_empty_ bucket hbw
burn ov
pick_up_and_ transport wt